<table>
<tr><td>Course Code</td><td>UE24CS342AA3</td><td>Course Title</td><td colspan="5">Internet of Things</td></tr>
<tr><td rowspan="2">Program</td><td rowspan="2">B.Tech</td><td rowspan="2">Hours per week/ Credit Assigned</td><td>L</td><td>T</td><td>P</td><td>S</td><td>C</td></tr>
<tr><td>4</td><td>0</td><td>0</td><td>4</td><td>4</td></tr>
<tr><td>Semester</td><td>5</td><td>Type of Course</td><td colspan="5">Elective-1</td></tr>
<tr><td>Tools/Languages</td><td></td><td>Desirable Knowledge</td><td colspan="5">Computer Networks</td></tr>
<tr><td>Prelude</td><td colspan="7">The Internet of Things (IoT) is no longer just about connecting devices—it is about building intelligent, autonomous, and secure cyber-physical systems. From smart homes and healthcare to industrial automation and agriculture, IoT systems operate at the intersection of embedded systems, networking, cloud computing, and security. This course provides a full-stack understanding of IoT, starting from the “things” (sensors, actuators, microcontrollers) and progressing through communication protocols, edge intelligence, cloud orchestration, and security fundamentals.</td></tr>
<tr><td>Course Objectives:</td><td colspan="7">● Understand the architecture, components, and functional stack of IoT systems.
● Develop embedded applications using microcontrollers and standard interfacing protocols.
● Analyze and apply IoT communication protocols for different use cases.
● Design data pipelines with edge computing and cloud integration for IoT applications.
● Identify and address security challenges in IoT systems, including authentication, encryption, and firmware risks.</td></tr>
<tr><td>Course Contents</td><td colspan="7">Unit 1 : Foundations – The "Things" and Embedded Logic.
Introduction to IoT, IoT Traffic Model, IoT Connectivity, IoT Architecture Basics: Perception (sensors), Network (gateways), and Application layers, IoTWF Model, The Core IoT Functional Stack.
The Things in IoT - Sensors, Actuators and smart Objects: Sensors - Its Types, Sensors & Its Performance Metrics, Sensor Selection, Actuators and Its Types MEMS, Smart Objects.
Microcontrollers (MCU): Deep dive into ESP32 and ARM Cortex-A, ARM Cortex-M architectures.
Embedded C & Python: Programming GPIOs, Interfacing (I2C, SPI, UART), and Interrupt handling.
14 Hours

Unit 2 : The Network – Connectivity & Protocols
Short-Range Protocols: BLE (Bluetooth Low Energy), Zigbee, and the new Matter/Thread standards (essential for 2026).
Long-Range Protocols: LoRaWAN and NB-IoT for industrial/agricultural use cases.
Application Protocols: MQTT (Publish/Subscribe), CoAP, and HTTP/WebSockets.
Lightweight Encryption: Standard Cryptography vs. Lightweight Introduction to ASCON (NIST standard) and how it differs from heavy AES.
Network Obfuscation: Hiding IoT devices behind gateways and using non-standard ports.
14 Hours

Unit 3 : Edge computing & Cloud Orchestration for IoT .
Introduction to IoT Analytics , IoT Data Pipelines & AI Integration,
Edge Computing: Introduction to TinyML—running machine learning models on microcontrollers.
Data Pipelines: Time-series databases (InfluxDB) and real-time visualization (Grafana).
Cloud Integration: Connecting to AWS IoT Core or Azure IoT Hub for device management.
Predictive Maintenance: Using sensor data to predict hardware failure before it happens.
14 Hours

Unit 4 : Introduction to IoT Security.
The IoT Attack Surface: Identifying vulnerabilities at each layer (Physical, Network, and Cloud).
Common Vulnerabilities (OWASP IoT Top 10): A beginner-friendly look at weak passwords, insecure network services, and lack of update mechanisms.
Zero Trust Architecture: Traditional network security vs Zero Trust, Limitations of perimeter-based security, Zero Trust principle: “Never Trust, Always Verify”, Types of device authentication methods.</td></tr>
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	Identity-based security vs network-based security. **Firmware Hacking :** Using **Binwalk** to look inside a firmware file for secrets. **X.509 certificates** - Introduction to X.509 certificate, Structure and components of digital certificates, Public key infrastructure (PKI) basics, Digital signatures and trust chain. **14 Hours**
Laboratory	Lab 1: Sensor Interfacing & Data Acquisition Lab 2: Communication Protocols Lab 3: Edge Processing & TinyML Lab 4: Cloud Integration & Visualization Lab 5: IoT Security & Firmware Analysis
Text Book(s):	1. IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things * Authors: David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton, and Jerome Henry (Cisco Press) 2. Designing the Internet of Things * Authors: Adrian McEwen and Hakim Cassimally.
Reference Book(s):	1. Advances in the Internet of Things: Challenges, Solutions, and Emerging Technologies 2. Internet of Things Security Attacks, Tools, Techniques and Challenges Preeti Mishra, Senthil Kumar Jagatheesa perumal,
Course Outcome	● Explain IoT architecture, components, and system-level design. ● Develop embedded IoT applications using microcontrollers and communication interfaces. ● Select and implement appropriate IoT communication protocols for given scenarios. ● Design IoT data pipelines with edge processing and cloud integration. ● Analyze IoT security threats and apply basic mechanisms for secure system design